

CliniScan – Chest X-ray Lung Abnormality Detection System

Milestone 2: Model Development & Baseline Training

Name: Vishnudharsh V V

Batch: B13

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1. Project Overview & Objective

The objective of Milestone 2 is to develop deep learning models capable of automatically analyzing chest X-ray images and identifying lung abnormalities.

The proposed system follows a two-stage pipeline consisting of a classification model and an object detection model.

The classification model determines whether a chest X-ray image is normal or abnormal, while the detection model

identifies the specific location of abnormalities using bounding boxes.

Pipeline Overview:

1. Classification Model – Predicts whether an X-ray image contains abnormalities.
2. Object Detection Model – If abnormalities are present, the model detects and localizes the affected regions.

2. Data Preparation Pipeline

2.1 Image Preprocessing

The dataset used for this project is a subset of the VinBigData Chest X-ray Abnormalities Detection dataset obtained from Kaggle.

Initially, the dataset consisted of medical images stored in DICOM format. These images were converted to JPG format

to simplify processing and reduce file size.

Preprocessing steps performed:

- Converted DICOM images to JPG format using pydicom and OpenCV
- Converted images to grayscale
- Resized images to 224×224 resolution
- Normalized pixel values
- Verified image integrity using PIL and Matplotlib

The preprocessing stage produced 5000 cleaned chest X-ray images, which were used for training and evaluation.

2.2 YOLO Annotation Conversion

Bounding box annotations were extracted from the dataset's train.csv file.

Rows labeled as "No finding" were removed because they do not contain bounding box annotations.

The remaining annotations were converted to YOLO format which represents bounding boxes using normalized coordinates.

YOLO label format:

class_id center_x center_y width height

All coordinate values were normalized between 0 and 1, ensuring compatibility with the YOLOv8 object detection framework.

Dataset structure:

dataset/

images/

train/

val/

test/

labels/

train/

val/

test/

Dataset split:

Training: 3500 images

Validation: 750 images

Testing: 750 images

3. Classification Model (EfficientNet-B0)

A classification model was developed to determine whether a chest X-ray image contains abnormalities.

Model Configuration:

Base Model: EfficientNet-B0

Input Size: 224×224

Loss Function: BCEWithLogitsLoss

Optimizer: Adam

Learning Rate: 0.0003

Epochs: 20

Performance evaluation metrics:

- F1 Score
- AUC-ROC
- Training and validation loss curves

The model successfully learned to differentiate between abnormal and normal chest X-ray images.

4. Object Detection Model (YOLOv8s)

For the detection stage, the YOLOv8s object detection model was used.

Model Configuration:

Model: YOLOv8s

Pretrained Weights: COCO

Image Size: 640

Epochs: 50

Batch Size: 8

Training Platform: Kaggle GPU

Training produced two weight files:

best.pt – Best performing model during training

last.pt – Final model after the last epoch

These weights can be used to perform inference on new chest X-ray images and generate bounding box predictions.

5. Challenges & Observations

Class Imbalance:

The dataset contained many normal chest X-ray images compared to abnormal ones, which affected model learning.

Small Abnormal Regions:

Certain abnormalities appear as small regions which are difficult for detection models to localize.

Hardware Limitations:

Training deep learning models locally requires GPU resources, so training was performed using Kaggle GPU.

6. Conclusion

Milestone 2 successfully implemented a working AI pipeline for chest X-ray analysis.

The classification model identifies abnormal scans, while the YOLOv8 object detection model localizes abnormal regions.

These models provide a foundation for further improvements such as dataset balancing, model optimization,

and improved detection accuracy in later milestones.